6 **(a)** **(i)** Using $A = A_0 e^{-\lambda t}$, and $A_0 = 3000$

at t = 2.0 years, A = 2700, $\lambda_1 = 0.0527$
at t = 6.0 years, A = 2220, $\lambda_2 = 0.0502$
at t = 8.0 years, A = 2020, $\lambda_3 = 0.0494$ [1, correct calculation of any 1 λ]

Therefore, average λ = 0.0508
[1, using <u>at least 3 points</u> to determine λ]

Hence, half-life = ln 2 / 0.0508 [1]
= 13.7
= 14 years

(ii) Number of Plutonium-241 nuclei : Number of Americium-241 nuclei = 3 : 1

This means **¼ of Plutonium-241 nuclei has decayed** and changed to Americium-241 nuclei. **¾ of Plutonium-241 nuclei is remaining**.

$\frac{3}{4} N_0 = N_0\, e^{-\lambda t}$ [1]
t = 5.6 years (using λ = 0.051) or 5.8 years (using $t_{1/2}$ = 14 years) [1]

(b) Number of nuclide = 0.0010 / 60u = 1.004×10^{22}

Using A = λN, $A = \left(\frac{\ln 2}{5.27 \times 365 \times 24 \times 3600}\right)(1.004 \times 10^{22})$
$= 4.187 \times 10^{13}$ [1]

Power of radiation = (activity)(energy emitted per activity)
$= (4.187 \times 10^{13})(1.17 + 1.33)(10^{6})(1.6 \times 10^{-19})$
= 16.75 W [1]

Intensity = Power / Area = 16.75 / $4\pi(1.5)^2$ = 0.592 W m^{-2} [1]

(Note: ECF for 2nd and 3rd marks)

(c) 'random' here means:

- it is impossible to know when a nucleus will disintegrate/which nucleus will disintegrate next, [1]
- even though the <u>probability of decay per unit time of a nucleus</u> is <u>constant</u>. [1]